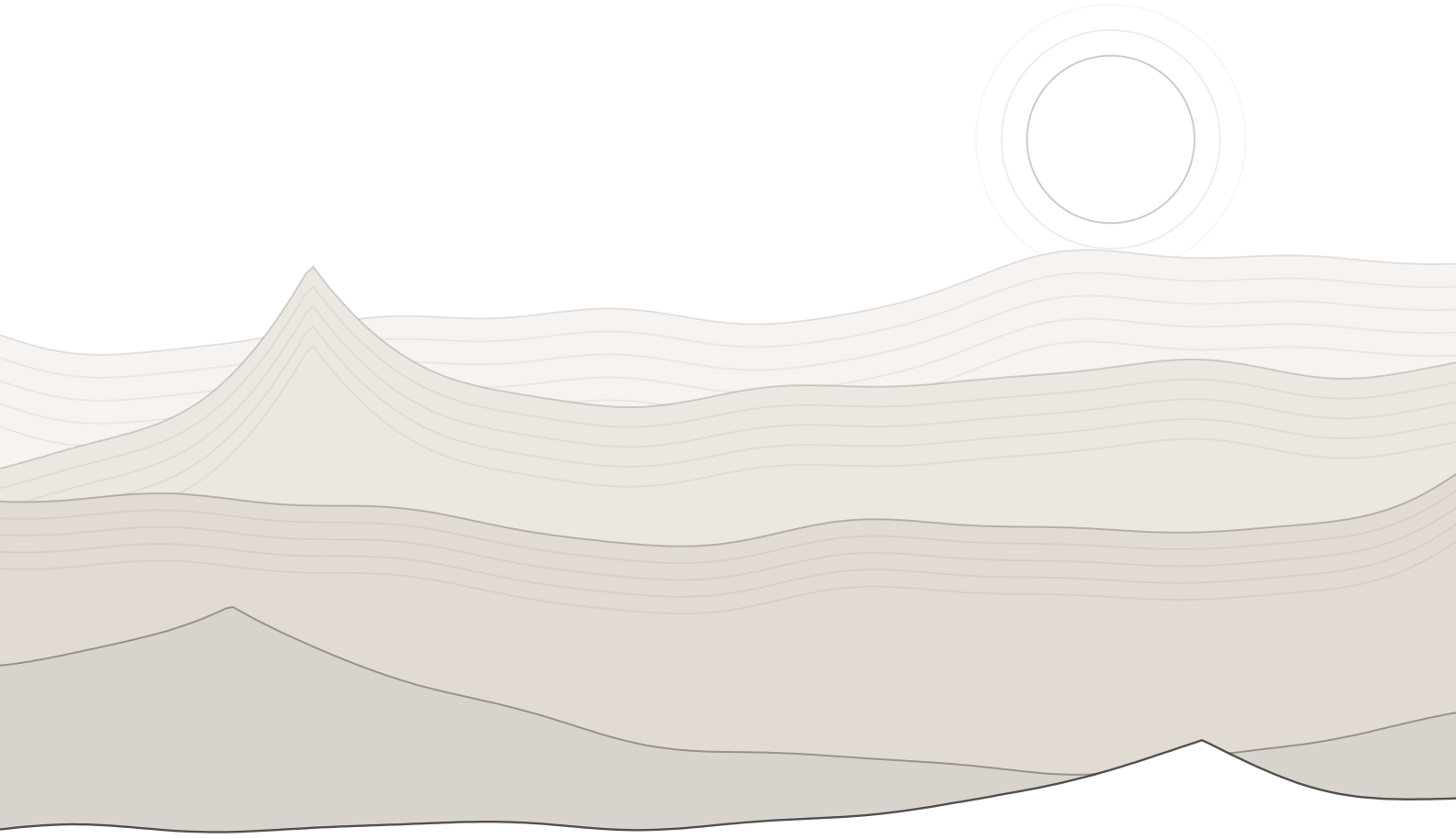
AMC 2026 · ROUND 1 · GROUP 3

# Strategic Asset Allocation

Part A, the analysis of the allocation: outlook, universe, inputs, method, allocation and backtest (pages 3–14), code and portfolio (page 17). Part B, the checkpoint assignments: investment policy (page 2), robust approaches and backtest protocol (pages 15–16, before the code).



| STARTING NAV | VOLATILITY TARGET | PUBLISHED |
|---|---|---|
| USD 100M | 8% p.a. | 18.10.26 |

CHECKPOINT 1

# Investment policy

## CP1.1 · UNIVERSE

Long-only US-listed ETFs on broad asset classes, in USD; no single stocks, shorts or leverage; cash earns nothing. Eight ETFs in three baskets, one per macro regime (TIP also partly hedges inflation); BIL only above the 9% cap.

| EQUITY · GROWTH | BONDS · DISINFLATION | REAL ASSETS · INFLATION | CASH LEG · ABOVE 9% |
|---|---|---|---|
| VTI · VEA · IEMG | BND · BNDX · TIP | GLD · BCI | BIL |

## CP1.2 · OBJECTIVE AND RISK BUDGET

Objective: the highest Sharpe Ratio over the 1-month T-bill, holding up in all four macro regimes (section 01).

| VOL. TARGET | EX-ANTE VOL. | 1-MONTH 5% VAR | MAX DRAWDOWN |
|---|---|---|---|
| 8% | 8.13% | 3.86% | −15% |
| Range 7–9% p.a. | Last 60 months | = USD 3.86 million | Acceptable · backtest −13.0% |

Worst loss of our weights since 2013: −13.0% (2022). Beyond −15% we would review the allocation at Round 2.

## CP1.3 · CONSTRAINTS, WITH REASONS

| CONSTRAINT WE IMPOSE | REASON (TODAY'S MODEL ERRORS) |
|---|---|
| **Concentration: risk, not capital.** Set first: equity ≤ 70% of the risk, bonds + commodities ≥ 30% (split equally), equal risk per ETF in a basket, no ETF above 20% of capital or risk. We chose 50/25/25 of four budgets (section 05). | Estimation error: on five years the mean of a 15%-vol ETF has a standard error of about 7 pp, so expected returns cannot set weights (an optimiser on them holds 35% in gold). Allocating by ETF lets the ETF count decide. |
| **Regions.** Each equity region at least 5% of the capital, for full coverage; equity weighted by risk, not market cap (35 / 33 / 33 of equity, against 64% US in the MSCI ACWI [6]); non-US bonds hedged to USD. | Model error, concentration: VTI's top ten are 36.1% of the fund [5], mostly AI mega caps whose earnings partly finance each other (chipmakers fund AI labs that buy their chips and cloud [17]): one factor the covariance misses. FX risk is unrewarded. |
| **Liquidity.** Fund assets above USD 1 billion; each position bought within one day of average volume; whole shares; no position below 2%. | Model error: the optimiser assumes trading at the close at no cost. BCI, our smallest fund (USD 3.5 billion), trades ≥ 1.5 million shares a day (USD 39.6 million); our USD 12.4 million is 31% of a day. Other seven: checked on Bloomberg, all pass. |
| **Cash leg.** BIL only above the 9% cap (risky weights scaled down to 9%). Our SAA (8.13%) holds none; the optimiser's 0.01% residual goes to equity (section 09). | Cash earns 0, about −4% a year against the 1-month T-bill (4.13%); BIL earns about the T-bill. |

## CP1.4 · RISK CONTRIBUTIONS

| ETF | WEIGHT | RISK CONTR. | ETF | WEIGHT | RISK CONTR. |
|---|---|---|---|---|---|
| VTI | 10.90% | 15.21% | TIP | 15.26% | 8.84% |
| VEA | 10.24% | 17.42% | GLD | 11.15% | 15.49% |
| IEMG | 10.23% | 17.36% | BCI | 12.40% | 9.51% |
| BND | 13.33% | 8.34% | BIL | 0.01% | 0.00% |
| BNDX | 16.48% | 7.83% | Total | 100.00% | 100.00% |

**Position above half of the risk: none** (largest: VEA, 17.4%). Equity holds 31.4% of capital and 50.0% of risk; bonds 45.1% and 25.0%; real assets 23.5% and 25.0%. Equity is the growth engine (its premium rests on growth and earnings); bonds and commodities brake in falling and rising inflation. Bound: 70% (60/40: ~90%, Class 3).

# Market outlook

Our reading of the macro picture is that 2026 has been a year of stagflation, that economists expect a return to goldilocks in 2027, and that markets do not share this optimism. Fed funds futures, in fact, price about 3.4 further hikes, reaching roughly 100 basis points above what economists forecast. Hence, we do not build the portfolio around a single regime, but hold assets that perform in each of them.

### 1.1 · GROWTH AND INFLATION

We classify each economy into the four regimes defined by the direction of growth and inflation, using the Bloomberg consensus forecasts. During 2026 inflation rose in every region, in the United States and in the emerging economies it did so while growth was slowing: US CPI went from 2.7% to 3.4% and real GDP growth fell from 2.6% to 1.8%, which places both economies in stagflation. For 2027, the consensus expects growth to remain steady or to rise and inflation to fall in every region, the pattern of a goldilocks economy. [1]

TABLE 1 · CONSENSUS FORECASTS AND MACRO REGIMES

| REGION | GDP Q4 26 | GDP Q4 27 | CPI Q4 26 | CPI Q4 27 | REGIME 2026 |
|---|---|---|---|---|---|
| United States | 2.1% | 2.1% | 3.5% | 2.1% | Stagflation |
| Developed economies | 1.8% | 1.9% | 3.9% | 2.5% | Overheated |
| Emerging economies | 3.9% | 4.4% | 3.6% | 2.9% | Stagflation |

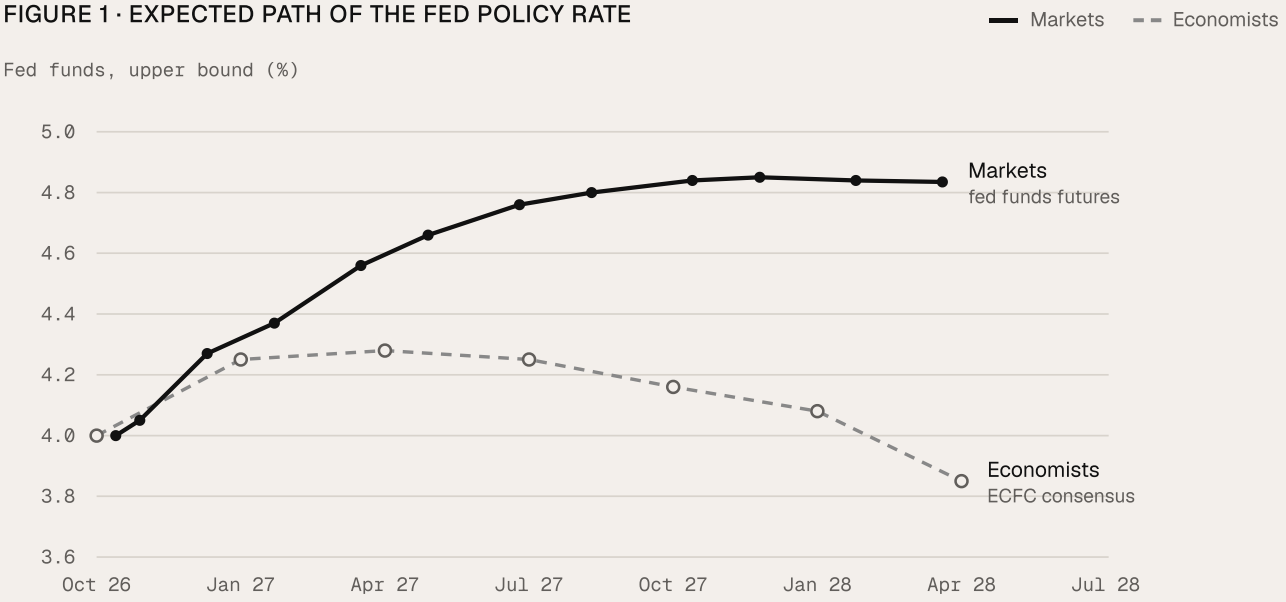
*Real GDP and CPI, year-on-year. Regime 2026: direction of growth and inflation from Q1 to Q3 2026. [1]*

Behind these numbers, the US economy is slowing gently rather than contracting. Unemployment is stable at about 4.2% and payroll growth has eased to about 60,000 jobs a month, while private investment remains the strongest component of demand, having grown at an annualised 7.4% in the third quarter. Inflation is still the main concern, with core PCE at 3.3%. Europe continues to grow slowly, at 1.3–1.6%, whereas the emerging economies combine rising growth with monetary easing, as their central banks are expected to cut rates from 6.3% to about 5.0% by early 2028. [1]

# Monetary policy: economists versus markets

After the September hike the Fed funds target range stands at 3.75–4.00%, and the path of the policy rate is where economists and markets disagree most. Economists expect one further hike followed by cuts that would bring the rate to 3.85% by early 2028, while fed funds futures price about 3.4 additional hikes, to an upper bound of about 4.84% by the end of 2027. [1, 2]

FIGURE 1 · EXPECTED PATH OF THE FED POLICY RATE




*Markets: implied upper bound from fed funds futures. Economists: Bloomberg ECFC consensus. [1, 2]*

IF ECONOMISTS ARE RIGHT

Bonds rally and growth assets benefit from lower rates.

IF MARKETS ARE RIGHT

Inflation remains sticky and both bonds and equities come under pressure.

We read this gap as the clearest evidence that the next regime is genuinely uncertain, and it is the starting point of our allocation.

1.3 · RATES, INFLATION PRICING, CREDIT AND THE DOLLAR

| INDICATOR | 1 YEAR AGO | LATEST |
|---|---|---|
| US Treasury 1-month yield | 4.20% | 4.13% |
| US Treasury 2-year yield | 3.60% | 4.80% |
| US Treasury 10-year yield | 4.14% | 5.24% |
| US Treasury 30-year yield | 4.72% | 5.60% |
| 5-year breakeven inflation | 2.39%* | 2.34% |
| 10-year real yield (TIPS) | 1.78% | 2.87% |
| US investment-grade spread (OAS) | 0.75% | 0.82% |
| US high-yield spread (OAS) | 2.84% | 3.15% |
| VIX | 16.3 | 15.4 |
| EUR/USD, consensus Q3 2026 → Q4 2027 | 1.13 | 1.17 |

*Table 2 · Key market indicators. Treasury yields and 1-year-ago real yield (8 Oct 2025): U.S. Treasury par and real yield curves [3]. Latest breakeven, real yield, spreads and VIX: FRED (T5YIE, DFII10, BAMLC0A0CM, BAMLH0A0HYM2, VIXCLS) [4]. *1-year-ago breakeven (9 Oct 2025) calculated as 5Y nominal 3.74% − 5Y real 1.35%. 1-year-ago VIX: close of 8 Oct 2025 [12]. EUR/USD [1].*

The indicators in Table 2 point to four features of the current market that shape our universe. The Treasury curve has shifted up by 80–130 basis points from the one-year maturity onwards, so that bonds once again offer a meaningful income. The inflation priced by the market, 2.34% over five years, is lower than the consensus forecast of about 2.8% for the next four quarters, which makes inflation protection inexpensive. Over the year the 10-year real yield rose by about 109 bp (1.78% → 2.87%) while five-year inflation expectations were almost unchanged (2.39% → 2.34%): the rise in nominal yields came from real rates, not from inflation expectations. Credit spreads remain tight, so that credit offers little compensation for its risk, and the consensus expects a mild weakening of the dollar, which would support non-US assets measured in USD.

# How we built the universe

**01**

**No bet on a single regime**

Given the outlook above, we hold assets that perform in each of the four regimes rather than picking one.

**02**

**Cheap, broad market exposures**

The policy portfolio drives most of the risk and return, so we buy beta passively: broad market-cap indices at low cost. Sector and factor bets are left for the tactical rounds.

**03**

**No duplicates**

No two ETFs track the same market. Two small overlaps remain, South Korea and gold, explained in section 03.

| EQUITY | FIXED INCOME | REAL ASSETS | RISK TOOL |
|---|---|---|---|
| VTI · VEA · IEMG | BND · BNDX · TIP | GLD · BCI | BIL |

*Eight strategic ETFs in three baskets, plus BIL, held only as a risk-management tool.*

# The ETFs and why we chose them

3.1 · EQUITY

Equity carries the equity risk premium, the most reliable source of long-run excess return. We hold three regional ETFs weighted by risk, not by market capitalisation. A market-cap allocation would be concentrated: 64% in the US (MSCI ACWI [6]), and much of it in a few AI-linked mega caps, as VTI's top ten holdings are 36.1% of the fund [5]. Weighting by risk hedges most of this out: the US is 35% of our equity.

## VTI Vanguard Total Stock Market ETF

US equity · Morningstar US Total Market Index · listed May 2001 · 3,519 stocks · TER 0.03% [5]

| | |
|---|---|
| WHY WE CHOSE IT | The deepest and most innovative equity market, with the strongest earnings growth of the last decade, led by technology and AI: our main source of expected return. |
| WHY NOT SPY / VOO | SPY and VOO hold the 500 largest companies; VTI adds mid and small caps, more sensitive to domestic growth and falling rates, and is slightly less concentrated in mega caps. |
| RISKS WE ACCEPT | High valuations and mega-cap concentration; with the 10-year yield at 5.24%, the equity risk premium is thin. The 20% cap per ETF and the two non-US ETFs limit both. |

## VEA Vanguard FTSE Developed Markets ETF

Developed ex-US equity · FTSE Developed All Cap ex US Index · listed July 2007

| | |
|---|---|
| WHY WE CHOSE IT | Developed ex-US markets trade at a discount to the US and hold more financials, industrials and healthcare and less technology, so they react differently to the same shocks. Unhedged, it gains from the expected weaker dollar (EUR/USD +3.5% to end-2027). |
| WHY NOT EFA / VXUS | EFA excludes Canada and small caps, which VEA includes; VXUS would merge in emerging markets, which we size separately. |
| RISKS WE ACCEPT | With European growth at 1.3–1.6%, the case rests on valuations and may take time. Exporters are exposed to tariffs, and a stronger dollar would cut the fund's USD value. |

## IEMG iShares Core MSCI Emerging Markets ETF

Emerging markets equity · MSCI Emerging Markets Investable Market Index · listed October 2012

| | |
|---|---|
| WHY WE CHOSE IT | The best macro mix of the three regions: growth rising from 3.9% to 4.4% in 2027, falling inflation, about 140 bp of rate cuts [1] and current-account surpluses of about 3% of GDP. Valuations are lower, and the region holds the chip supply chain (Taiwan, Korea). |
| WHY NOT EEM | IEMG includes small caps (IMI index) and is cheaper. |
| RISKS WE ACCEPT | Higher volatility, political risk in China, and losses when the dollar and US rates rise. South Korea is held twice: Developed for FTSE since 2009 [7], so in VEA, but still Emerging for MSCI in June 2026 [8]. |

### 3.2 · FIXED INCOME

Bonds lower the volatility enough to stay within the 7–9% risk budget and protect the portfolio in a recession with falling inflation. We hold three sources of return: US nominal rates, non-US rates and US real rates.

## BND Vanguard Total Bond Market ETF

US investment-grade bonds · Bloomberg US Aggregate Float Adjusted Index · listed April 2007

| | |
|---|---|
| WHY WE CHOSE IT | Treasuries, agency MBS and IG corporate bonds at intermediate duration. With the 10-year yield at 5.24%, bonds pay a meaningful income, and the expected fall to about 4.6% by end-2027 [1] would add a capital gain. Above all, BND is our recession insurance. |
| WHY NOT TLT | With a deficit of about 6.5% of GDP and debt at 105% [1], Treasury supply may keep long yields high: intermediate duration keeps most of the hedge with less of this risk. High yield pays too little (spread 3.15%) for a risk that rises when equities fall. |
| RISKS WE ACCEPT | Since 2021 the stock–bond correlation has been positive: when inflation is the problem, both can fall, as in 2022. Futures also price further Fed hikes. |

## BNDX Vanguard Total International Bond ETF

International IG bonds, USD-hedged · Bloomberg Global Aggregate ex-USD Float Adjusted RIC Capped Index · listed June 2013

| | |
|---|---|
| WHY WE CHOSE IT | Rate cycles outside the US are not synchronised with the Fed: the ECB rate is about 2.85%, expected at 2.5% by early 2028 [1]. BNDX therefore diversifies the duration risk of BND, and with US short rates higher, its USD hedge also earns the rate differential. |
| WHY NOT IGOV | IGOV is unhedged: currency moves would add equity-like volatility to our defensive assets. |
| RISKS WE ACCEPT | A global inflation shock would push yields up everywhere at once. |

## TIP iShares TIPS Bond ETF

US inflation-linked Treasuries · listed December 2003

| | |
|---|---|
| WHY WE CHOSE IT | Its risk is mostly real-rate duration, which is why it sits in the bond basket: its correlation with BND is 0.90 (Table 4). It also partly hedges inflation, and cheaply: the 5-year breakeven (2.34%) is below the consensus CPI for the next four quarters (2.8%) [1, 4], and the 2.87% real yield pays even if inflation returns to target. |
| WHY NOT STIP | STIP holds only 0–5 year maturities; with real yields at their highest in years, we want some real-rate duration. |
| RISKS WE ACCEPT | TIP loses when real yields rise, even with high inflation, as in 2022; and if inflation falls faster than priced, it lags nominal bonds such as BND. |

### 3.3 · REAL ASSETS

Real assets protect the portfolio in the two regimes in which equities and nominal bonds can fall together — high inflation and supply shocks — and are the least correlated part of the universe.

## GLD SPDR Gold Shares

Gold · physical gold bullion · listed November 2004

| | |
|---|---|
| WHY WE CHOSE IT | Gold has no credit or counterparty risk and has held its value in stagflation, geopolitical crises and dollar weakness; central banks bought about 850 tonnes in 2025 [9]. With high US public debt and a weaker dollar expected, it is our insurance. |
| WHY NOT IAU | GLD is the largest and most liquid gold ETF, with the longest history; IAU is cheaper, but for a position held a few months liquidity matters more than fees. |
| RISKS WE ACCEPT | No income, with real yields near 2.9%, and a strong rally behind it: the LBMA price averaged a record USD 4,873/oz in Q1 2026, when central-bank buying slowed to 57 tonnes [9]. |
| OVERLAP WITH BCI | Gold is also 14.9% of the Bloomberg Commodity Index in 2026 (precious metals 18.8%, energy 29.4%) [18], so our gold is about 13.0% of the capital: 11.15% in GLD plus 1.8 pp through BCI. We accept this: investors treat gold as a standing asset class in inflation and stress, and the overlap adds less than 2 pp. |

## BCI Bloomberg All Commodity Strategy K-1 Free ETF

Broad commodities · Bloomberg Commodity Index Total Return · listed March 2017

| | |
|---|---|
| WHY WE CHOSE IT | Commodities gain when inflation comes from supply shocks, the case in which stocks and bonds both lose, as in 2022. Gold reacts to monetary and geopolitical stress, commodities to the real economy; the total-return index also earns the T-bill rate on the collateral. |
| WHY NOT DBC | BCI follows the Bloomberg Commodity Index: 23 commodities, no sector above 33% and no single commodity above 15% [11]. DBC is about half energy and more expensive [10]. |
| RISKS WE ACCEPT | Roll costs in markets in contango, and sharp falls in a recession, when demand for raw materials collapses. |

## BIL SPDR Bloomberg 1-3 Month T-Bill ETF

3.4 Risk-management tool · US Treasury bills, 1-3 months · listed May 2007

| | |
|---|---|
| ROLE | Not a strategic position. BIL is used only if the risk budgets give an ex-ante volatility above the 9% cap: the risky weights are scaled down to 9% and the rest goes to BIL, as in the 70/15/15 budget (Table 5). Our SAA, at 8.13%, holds none: the optimiser's 0.01% residual goes to equity (section 09). |
| WHY NOT CASH | The Sharpe Ratio is measured on returns in excess of the 1-month T-bill (4.13%) [3]. Cash earns nothing, so its excess return is about –4% a year; BIL's is close to zero. |
| WHY NOT SGOV | SGOV is equivalent and slightly cheaper, but BIL's longer record (2007, against 2020) gives more data for the backtest. |

# The inputs

All inputs are estimated on the same 60 complete months, **October 2021 – September 2026**, as the guidelines require. Returns are monthly total returns in USD, computed from Bloomberg adjusted prices, which include dividends [13]. Means and variances are annualised ×12, volatilities ×$\sqrt{12}$. BCI only started trading in 2017, so for the whole sample we use the index it tracks, the Bloomberg Commodity Index Total Return (BCOMTR), before the fund's fees. In this way, the estimation window and the backtest use the same series. The risk-free rate is the 1-month T-bill (FRED DGS1MO) [14], the rate on which the Sharpe Ratio of the challenge is computed.

TABLE 3 · VOLATILITY AND MEAN

| ETF | VOLATILITY | HIST. EXCESS RETURN* |
|---|---|---|
| VTI | 15.24% | 8.97% |
| VEA | 15.67% | 7.02% |
| IEMG | 16.28% | 5.86% |
| BND | 6.76% | −4.25% |
| BNDX | 5.57% | −3.74% |
| TIP | 6.00% | −3.87% |
| GLD | 17.08% | 14.58% |
| BCI† | 15.36% | 8.10% |

**Mean monthly return minus the T-bill, ×12. Shown, not used by our method. †BCOMTR index.*

TABLE 4 · CORRELATIONS

| | VTI | VEA | IEMG | BND | BNDX | TIP | GLD | BCI |
|---|---|---|---|---|---|---|---|---|
| VTI | 1 | 0.82 | 0.72 | 0.50 | 0.53 | 0.53 | 0.14 | 0.09 |
| VEA | 0.82 | 1 | 0.81 | 0.64 | 0.58 | 0.62 | 0.42 | 0.20 |
| IEMG | 0.72 | 0.81 | 1 | 0.56 | 0.47 | 0.53 | 0.44 | 0.24 |
| BND | 0.50 | 0.64 | 0.56 | 1 | 0.89 | 0.90 | 0.46 | −0.06 |
| BNDX | 0.53 | 0.58 | 0.47 | 0.89 | 1 | 0.85 | 0.41 | −0.13 |
| TIP | 0.53 | 0.62 | 0.53 | 0.90 | 0.85 | 1 | 0.47 | 0.12 |
| GLD | 0.14 | 0.42 | 0.44 | 0.46 | 0.41 | 0.47 | 1 | 0.31 |
| BCI | 0.09 | 0.20 | 0.24 | −0.06 | −0.13 | 0.12 | 0.31 | 1 |

*−1 +1: blue to red through white at 0, as in the notebook. Monthly returns, Oct 2021 – Sep 2026.*

### 4.1 · WHAT THE CORRELATIONS TELL US

**Each basket behaves almost as one asset.** The three equity ETFs correlate 0.72–0.82 and the three bond ETFs 0.85–0.90, so there is little diversification inside a basket. Diversification comes from combining the baskets, which is why our method sets the risk of each basket (section 05).

**Stocks and bonds have moved together.** Their correlation is 0.47–0.64 over a window that includes 2022. Bonds still lower the volatility, but they hedge equities less than they did before 2021 (Class 3, Part II). For this reason we do not rely on bonds alone for protection.

**Gold and commodities are the real diversifiers.** Their correlations with the other ETFs are at most 0.47, and commodities are uncorrelated with bonds (−0.13 to 0.12). This is the statistical side of the role that we give to real assets in section 03.

### 4.2 · WHY WE DO NOT USE THE EXPECTED RETURNS

The historical excess returns describe the last five years, not the next ones. Bonds show −4% a year because yields rose sharply in 2022, the same rise that now gives bonds a yield above 5%. Gold shows +14.6% because of a record rally. With five years of data, the standard error of the mean of an ETF with a 15% volatility is 15% / $\sqrt{5}$ ≈ 6.7 percentage points, larger than most of the means themselves. An optimiser using these numbers puts a third of the capital in gold and nothing in bonds (section 05). Our weights therefore depend only on the volatilities and the correlations, which are estimated much more precisely.

# The method: risk budgets by basket

Our weights come from a two-level risk-budgeting rule that uses only the covariance matrix: equal risk per ETF inside each basket (ERC) and, across the baskets, **50% equity, 25% bonds, 25% real assets**. Long-only, no leverage, ex-ante volatility between 7% and 9%.

## 5.1 · HOW THE WEIGHTS ARE COMPUTED

**Step 1 · inside the basket**

ERC weights, so that $w_i(Vw)_i$ / w′Vw is the same for every ETF of the basket. Unlike 1/σ, ERC uses the correlations.

**Step 2 · across baskets**

Each basket becomes one return series. On their covariance matrix we solve min $\Sigma_k(\%RC_k - b_k)^2$, with b = 50 / 25 / 25, w ≥ 0, Σw ≤ 1 and 7% ≤ σ ≤ 9%.

**Step 3 · combine**

ETF weight = basket weight × weight inside the basket; BIL takes the rest only when the 9% cap binds. Solved numerically (SLSQP, scipy).

## 5.2 · WHY A RISK-BASED METHOD

**The asset class, not the ETF count.** Equal weight, risk parity 1/σ and ERC on single ETFs are asset-class agnostic: they allocate by ETF, so the result depends on how many ETFs each asset class has (under ERC, three bond ETFs out of eight take 3/8 of the risk). Our method budgets the risk by asset class.

**Estimation error.** Against mean–variance and maximum Sharpe the reason is estimation error: small errors in expected returns become extreme weights. In Class 3's 40-year horse race, mean–variance earned an out-of-sample Sharpe Ratio of 0.04, against 0.53 for risk parity; our historical means are noisy too (section 04).

**Regimes; capital is not risk.** Equities pay in goldilocks, bonds in a recession with falling inflation, real assets under inflation (Class 3): fixing the risk of each basket means no regime dominates, and the next one is uncertain (section 01). Budgeting risk, not capital, controls what each basket can do to the portfolio.

## 5.3 · WHY 50 / 25 / 25

We allocate risk, not capital. Equity is the growth engine: its premium, the best-documented long-run premium, rests on growth and earnings [16]; bonds and commodities are the brakes, in falling and rising inflation. Equal risk across baskets is optimal only with equal Sharpe Ratios and correlations [15], so we set the rules before the weights: equity gets the largest share of the risk, at most 70%; the brakes at least 30%, split equally.

Within these rules we tested four budgets (Table 5) and chose 50/25/25 as the closest to the 8% target: the effect of a risk budget on the volatility is hard to anticipate ex ante. The backtests only confirmed practically the same hold-out Sharpe Ratio (0.65–0.66). A 60/40 portfolio has about 90% of its risk in equity: even our 70% budget is considerably more balanced.

TABLE 5 · THE FOUR EQUITY BUDGETS

| RISK BUDGET EQUITY / BONDS / REAL | EX-ANTE VOL. | BIL | SHARPE, HOLD-OUT |
|---|---|---|---|
| 33 / 33 / 33 | 7.39% | 0.06% | 0.65 |
| **50 / 25 / 25 · our SAA** | **8.13%** | **0.01%** | **0.66** |
| 60 / 20 / 20 | 8.71% | 0.00% | 0.66 |
| 70 / 15 / 15 | 9.00% | 4.71% | 0.66 |

*Hold-out: Jul 2013 – Sep 2021, months not used for the weights. At 70/15/15 the 9% cap binds and the rest goes to BIL. The BIL of up to 0.06% in the other budgets is a numerical residual (risk budgets fix the shares of risk, not the scale); for our SAA it goes to equity in section 09.*

# The methods we compared

We computed eleven allocations on the same inputs and rules (long-only, no leverage, rest in BIL, ex-ante volatility in 7–9%). Only the four basket budgets were candidates. Equal weight and risk parity 1/σ are the benchmarks of Checkpoint 2. The other five are diagnostics, never candidates, kept to show the estimation-error problem.

TABLE 6 · WEIGHTS OF THE ELEVEN ALLOCATIONS

| METHOD | VTI | VEA | IEMG | BND | BNDX | TIP | GLD | BCI | BIL | VOL. |
|---|---|---|---|---|---|---|---|---|---|---|
| CANDIDATES — BASKET METHOD | | | | | | | | | | |
| Basket 33/33/33 | 7.1 | 6.7 | 6.7 | 15.7 | 19.4 | 18.0 | 12.4 | 13.8 | 0.1 | 7.39 |
| **Basket 50/25/25 · SAA** | **10.9** | **10.2** | **10.2** | **13.3** | **16.5** | **15.3** | **11.1** | **12.4** | **0.0** | **8.13** |
| Basket 60/20/20 | 13.5 | 12.7 | 12.7 | 11.7 | 14.5 | 13.4 | 10.2 | 11.4 | 0.0 | 8.71 |
| Basket 70/15/15 | 15.8 | 14.8 | 14.8 | 9.4 | 11.6 | 10.7 | 8.6 | 9.5 | 4.7 | 9.00 |
| BENCHMARKS OF CHECKPOINT 2 | | | | | | | | | | |
| Equal weight | 12.5 | 12.5 | 12.5 | 12.5 | 12.5 | 12.5 | 12.5 | 12.5 | 0.0 | 8.78 |
| Risk parity 1/σ | 8.1 | 7.9 | 7.6 | 18.3 | 22.2 | 20.6 | 7.2 | 8.1 | 0.0 | 7.23 |
| DIAGNOSTICS — NEVER CANDIDATES | | | | | | | | | | |
| Minimum variance | 9.8 | 9.2 | 9.2 | 11.3 | 11.6 | 11.4 | 9.8 | 11.0 | 16.7 | 7.00 |
| Maximum Sharpe | 26.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 35.3 | 14.6 | 24.0 | 8.61 |
| Mean–variance at 8% | 24.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 32.9 | 13.5 | 29.5 | 8.00 |
| ERC, 8 ETFs | 8.2 | 6.7 | 6.8 | 16.7 | 21.8 | 17.9 | 8.0 | 13.7 | 0.2 | 7.13 |
| Max. diversification | 16.4 | 0.0 | 0.0 | 5.3 | 41.4 | 0.0 | 10.6 | 26.4 | 0.0 | 7.00 |

*Weights and ex-ante volatility in %, Oct 2021 – Sep 2026. Equal weight and 1/σ are fully invested by definition; the others are solved under the 7–9% band.*

## WHY WE DID NOT CHOOSE THE OTHERS

| | |
|---|---|
| OTHER CANDIDATES<br>**33/33/33, 60/20/20, 70/15/15** | Practically the same hold-out Sharpe Ratio as ours (Table 5), but further from the 8% target: 7.39%, 8.71% and 9.00%, the last only with 4.71% in BIL. Equal baskets give equity only a third of the risk. |
| BENCHMARKS<br>**Equal weight, risk parity 1/σ** | Asset-class agnostic: they allocate by ETF and let the ETF count decide. Equal weight ignores the risk of each ETF; 1/σ ignores the correlations. In the hold-out test, risk parity 1/σ outperformed our allocation (Sharpe 0.78 against 0.66), and we did not adopt it: we chose among the candidates on economic grounds. |
| DIAGNOSTICS<br>**Maximum Sharpe, mean–variance** | Estimation error: the only rules that use the expected returns, and it shows: a third of the capital in gold, a quarter in US equity, no bonds, no non-US equity and 24–30% in BIL. They load on the winners of the last five years (Class 3). |
| DIAGNOSTICS<br>**Minimum variance** | It stops at the 7% floor of the band and is not unique: every portfolio at exactly 7% is optimal, so the weights depend on where the solver starts. It also leaves 16.7% in BIL. |
| DIAGNOSTICS<br>**Maximum diversification** | It puts 41% in BNDX and 26% in commodities and drops non-US equity: a portfolio built on one estimated correlation structure. |
| DIAGNOSTICS<br>**ERC on the 8 ETFs** | The ETF count becomes a hidden risk budget: three bond ETFs out of eight receive 3/8 of the risk. It also stays near the bottom of the band (7.1%). |

# The allocation

| EX-ANTE VOLATILITY | 1-MONTH 5% VAR | POSITIONS | LARGEST RISK CONTR. |
|---|---|---|---|
| 8.13% | 3.86% | 8 ETFs | 17.4% |
| Band 7–9% | USD 3,860,465 | Three baskets · BIL not held | VEA |

TABLE 7 · CAPITAL WEIGHT AND RISK CONTRIBUTION

Capital Risk

| ETF | CAPITAL | RISK | |
|---|---|---|---|
| EQUITY · 31.37% OF CAPITAL · 50.0% OF RISK | | | |
| VTI | 10.90% | 15.21% | |
| VEA | 10.24% | 17.42% | |
| IEMG | 10.23% | 17.36% | |
| BONDS · 45.07% OF CAPITAL · 25.0% OF RISK | | | |
| BND | 13.33% | 8.34% | |
| BNDX | 16.48% | 7.83% | |
| TIP | 15.26% | 8.84% | |
| REAL ASSETS · 23.54% OF CAPITAL · 25.0% OF RISK | | | |
| GLD | 11.15% | 15.49% | |
| BCI | 12.40% | 9.51% | |
| BIL | 0.01% | 0.00% | Cash leg · residual, goes to equity (section 09) |

### 6.1 · CAPITAL VERSUS RISK

Bonds take 45% of the capital but only 25% of the risk; equity takes 31% of the capital and 50% of the risk. A dollar in equity moves the portfolio about 2.6 times as much as a dollar in bonds (volatility 15–16% against 6–7%). The capital weights therefore look defensive, but the risk is split evenly between growth and protection, by design.

### 6.2 · INSIDE THE BASKETS

VEA and IEMG get the same weight although IEMG is more volatile (16.3% against 15.7%), because IEMG is less correlated with VTI (0.72 against 0.82). In the whole portfolio, however, VEA and IEMG contribute more risk than VTI, because they are more correlated with bonds and gold. BNDX has the largest weight because it has the lowest volatility (5.6%). Between the two real assets, BCI gets more capital because it is less volatile; it also contributes less risk than gold (9.5% against 15.5%), because it is the least correlated asset of the portfolio.

### 6.3 · VOLATILITY AND VALUE AT RISK

The ex-ante volatility, $\sqrt{(w'Vw)} = 8.13\%$, is inside the band and close to the target. Under normality, the 1-month 5% VaR is $1.645 \times 8.13\% / \sqrt{12} = 3.86\%$: in one month out of twenty we expect to lose more than USD 3.86 million. We read this as a reference, not a ceiling. In the backtest the worst month of these weights was –9.4% (March 2020), and the monthly returns are skewed to the left, so the true tail is fatter than the normal one. Realised volatility can also differ from 8.1%: the window includes 2022, and in calmer years such as 2013–2021 the same weights had a volatility of 6.9%.

# The backtest

We hold the SAA weights fixed and rebalance them to target at the end of every month; whatever is not invested earns the BIL return. The sample is **July 2013 – September 2026** (159 months): it starts when every ETF exists, the limit being BNDX (June 2013). There are no transaction costs, since the guidelines do not mention them and monthly rebalancing only trades the drift of the weights. The measures are those of Class 2. Because the weights are estimated on the last 60 months, a single backtest would partly test the weights on the data that produced them. We therefore run three tests, each with its own question and limit:

**Full sample**

Our weights on all 159 months. Reference only: the last 60 months are the estimation window, so they are in-sample.

**Hold-out · the test we rely on**

Our weights on the 99 months before the estimation window, never seen by the optimiser. No in-sample fitting, but look-ahead (2021–2026 weights on 2013–2021). It tests the mix we hold, not the method or re-optimising.

**Split**

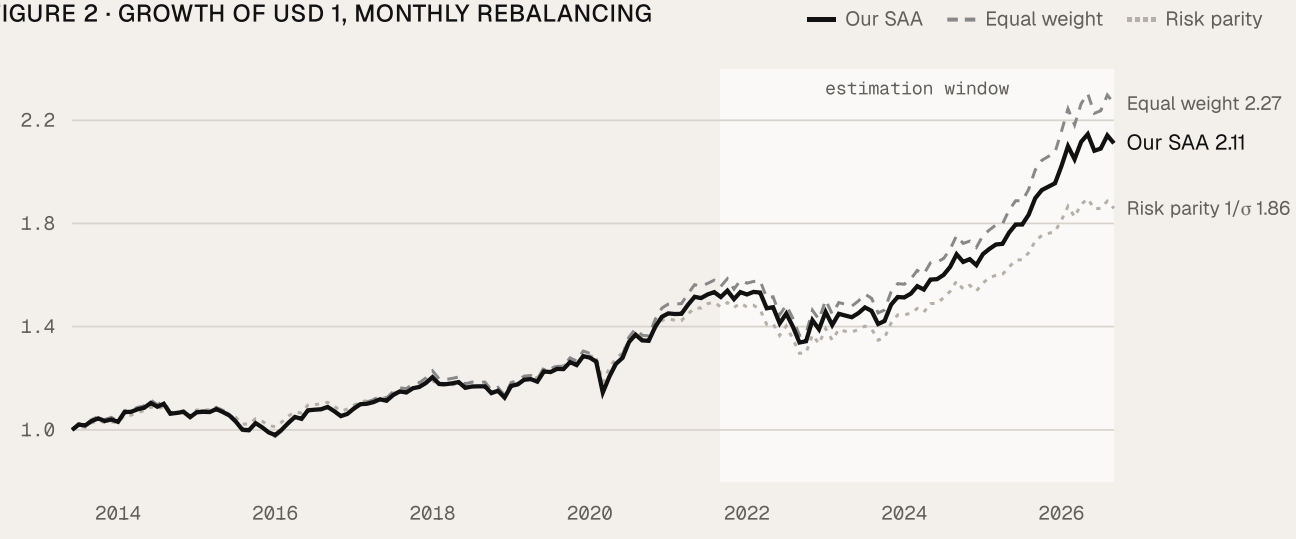
The rule estimated once on the first 60 months and held fixed on the following 99, as our SAA is held fixed. No look-ahead, but it tests the method: its weights are not ours.

TABLE 8 · PERFORMANCE AND RISK OF THE SAA

| | FULL SAMPLE | HOLD-OUT | SPLIT |
|---|---|---|---|
| Period | 07.13-09.26 | 07.13-09.21 | 07.18-09.26 |
| Return, annualised | 5.92% | 5.28% | 9.75% |
| Cumulative return | 111.1% | 51.5% | 113.2% |
| Volatility, annualised | 7.39% | 6.94% | 10.35% |
| Sharpe Ratio (t-stat.) | 0.55 (2.00) | 0.66 (1.91) | 0.68 (1.94) |
| Sortino Ratio | 0.83 | 1.01 | 1.01 |
| Maximum drawdown | -13.04% | -11.24% | -15.57% |
| 1-month 5% VaR | 3.51% | 3.30% | 4.92% |

*Sharpe Ratio on returns in excess of the 1-month T-bill; t-statistic of the mean excess return = Sharpe × √years. VaR = 1.645 × realised monthly volatility, as for the ex-ante VaR. Split tests the rule, with weights different from ours.*

FIGURE 2 · GROWTH OF USD 1, MONTHLY REBALANCING




*Full sample, Jul 2013 – Sep 2026, fixed weights estimated on Oct 2021 – Sep 2026 (light band). Equal weight and risk parity 1/σ are the comparisons of Checkpoint 2, not a benchmark. [13, 14]*

# What the backtest tells us

TABLE 9 · CALENDAR-YEAR RETURNS

| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Our SAA** | **0.8** | **-5.7** | **7.3** | **11.3** | **-4.8** | **14.3** | **11.9** | **6.7** | **-9.4** | **9.0** | **8.2** | **19.3** | **7.9** |
| Equal weight | 0.3 | -6.1 | 7.6 | 13.1 | -5.7 | 15.3 | 12.7 | 7.3 | -9.6 | 9.8 | 8.9 | 21.6 | 9.1 |
| Risk parity 1/σ | 2.5 | -3.8 | 6.3 | 9.1 | -3.4 | 12.7 | 10.9 | 4.7 | -10.6 | 8.4 | 6.3 | 15.0 | 5.1 |

*Total return in %, USD. *January – September 2026. Fixed weights of section 06 for our SAA.*

### 7.2 · READING THE RESULTS

**Return and risk.** Over the full sample, USD 1 grew to USD 2.11: 5.9% a year with 7.4% volatility and a Sharpe Ratio of 0.55. The portfolio lost money in three of the twelve full calendar years (2015, 2018, 2022), never more than 9.4%.

**2022, the stress test that matters.** From October 2021 to September 2022 the portfolio lost 13.0%, its maximum drawdown. In 2022 US equity fell 20% and US bonds 12.5%, so the usual hedge failed. Commodities (+15.7%) and gold (+0.8%) were the only positions that gained, which is the regime for which we hold them. The loss was recovered by March 2024. In March 2020 (–9.4% in the month, –10.9% from the peak) equities fell about 20% and commodities 17%, while bonds and gold held their value; the loss was recovered by July.

**Volatility depends on the regime.** With the same weights, realised volatility was 6.9% in 2013–2021, when stocks and bonds were negatively correlated, against 8.1% in the estimation window. Our weights are calibrated on the recent, less favourable regime, so they should not exceed the band if that regime persists.

**The evidence is statistically weak.** The t-statistics of the mean excess return are between 1.9 and 2.0, at the usual threshold of about 2: only the full sample, which is partly in-sample, just reaches it. This is the Class 2 lesson that decades of data are needed, and it is why Checkpoint 3 discounts the backtested Sharpe Ratio.

TABLE 10 · SHARPE RATIO AND EX-POST VOLATILITY OF THE ELEVEN ALLOCATIONS

| | FULL SAMPLE | | HOLD-OUT | |
|---|---|---|---|---|
| | SHARPE | VOL. | SHARPE | VOL. |
| CANDIDATES — BASKET METHOD | | | | |
| Basket 33/33/33 | 0.50 | 6.46% | 0.65 | 5.86% |
| **Basket 50/25/25 · SAA** | **0.55** | **7.39%** | **0.66** | **6.94%** |
| Basket 60/20/20 | 0.57 | 8.12% | 0.66 | 7.79% |
| Basket 70/15/15 | 0.58 | 8.62% | 0.66 | 8.41% |
| BENCHMARKS OF CHECKPOINT 2 | | | | |
| Equal weight | 0.57 | 8.17% | 0.64 | 7.81% |
| Risk parity 1/σ | 0.49 | 6.15% | 0.78 | 5.44% |

| | FULL SAMPLE | | HOLD-OUT | |
|---|---|---|---|---|
| | SHARPE | VOL. | SHARPE | VOL. |
| DIAGNOSTICS — NEVER CANDIDATES | | | | |
| Minimum variance | 0.55 | 6.45% | 0.64 | 6.11% |
| Maximum Sharpe | 0.78 | 8.19% | 0.64 | 7.85% |
| ERC, 8 ETFs | 0.49 | 6.25% | 0.68 | 5.69% |
| Mean–variance at 8% | 0.77 | 7.60% | 0.63 | 7.28% |
| Max. diversification | 0.56 | 6.57% | 0.60 | 6.31% |

*Ex-post volatility: realised, annualised. Full sample Jul 2013 – Sep 2026; hold-out Jul 2013 – Sep 2021.*

Maximum Sharpe leads the full sample (0.78) only because its weights were fitted on the last 60 months; in the hold-out it falls to 0.64. In the hold-out, risk parity 1/σ has the highest Sharpe Ratio (0.78) and the lowest volatility (5.44%). Our basket is never the best nor the worst (0.55 in the full sample, 0.66 in the hold-out), and the four candidates have practically the same hold-out Sharpe Ratio (0.65–0.66). The hold-out has look-ahead bias, but it is the test of the mix we hold, not of the method and not of re-optimising.

CHECKPOINT 2

# Our allocation vs. the robust approaches

## CP2.1 · WEIGHTS, RISK CONTRIBUTIONS AND VOLATILITY

EQUAL WEIGHT

$$w_i = 1 / N$$

RISK PARITY

$$w_i \propto 1 / \sigma_i$$

Both rules are applied to our eight ETFs, fully invested. The volatilities for risk parity, the risk contributions and the volatility of each portfolio use the same 60 months as our covariance matrix (October 2021 – September 2026).

| | OUR SAA | | EQUAL WEIGHT | | RISK PARITY | |
|---|---|---|---|---|---|---|
| ETF | WEIGHT | RISK CONTR. | WEIGHT | RISK CONTR. | WEIGHT | RISK CONTR. |
| VTI | 10.90% | 15.21% | 12.50% | 16.43% | 8.11% | 12.69% |
| VEA | 10.24% | 17.42% | 12.50% | 19.97% | 7.89% | 14.90% |
| IEMG | 10.23% | 17.36% | 12.50% | 20.05% | 7.59% | 13.98% |
| BND | 13.33% | 8.34% | 12.50% | 6.98% | 18.29% | 14.35% |
| BNDX | 16.48% | 7.83% | 12.50% | 5.26% | 22.20% | 13.47% |
| TIP | 15.26% | 8.84% | 12.50% | 6.45% | 20.62% | 14.69% |
| GLD | 11.15% | 15.49% | 12.50% | 15.96% | 7.24% | 10.73% |
| BCI | 12.40% | 9.51% | 12.50% | 8.91% | 8.05% | 5.18% |
| BIL | 0.01% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% |
| Equity | 31.4% | 50.0% | 37.5% | 56.4% | 23.6% | 41.6% |
| Bonds | 45.1% | 25.0% | 37.5% | 18.7% | 61.1% | 42.5% |
| Real assets | 23.5% | 25.0% | 25.0% | 24.9% | 15.3% | 15.9% |
| Volatility | 8.13% | | 8.78% | | 7.23% | |

## CP2.2 · WHERE THE RULES DISAGREE WITH OUR WEIGHTS

The three allocations share the same shape: every ETF holds 7–22% of the capital, all within the 7–9% band. They disagree in three places. **Equity:** equal weight gives it 37.5% of the capital and 56% of the risk, because it ignores that equities are 2.6 times as volatile as bonds; risk parity gives it 24% and 42%. Our 31% of capital and exactly 50% of risk lies in between, and it is the one choice in our weights that rests on a view about returns. That view is a prior that the equity premium is the most reliable premium, not an estimated mean. **Bonds:** risk parity puts 61% of the capital in the three bond ETFs because each looks safe on its own. But they are correlated 0.85–0.90 with each other and, since 2021, positively with equities, so $1/\sigma$ counts the same risk three times. Our ERC uses the correlations and holds 45%. **Real assets:** risk parity cuts gold and commodities to 15% of the capital and 16% of the risk because they are volatile on their own. It ignores that they are the least correlated assets in the universe and the only ones that gained in 2022. We keep them at 25% of the risk, on the basis of correlations and our regime analysis, not their past returns. None of our weights depends on an estimated expected return. The positions that would — 33–35% in gold under maximum Sharpe and mean–variance — are exactly the ones we rejected. In the hold-out test, risk parity $1/\sigma$ has a higher Sharpe Ratio than our allocation (0.78 against 0.66), and equal weight a similar one (0.64). Our weights are therefore justified on economic grounds, not by superior backtested performance.

CHECKPOINT 3

# Backtest protocol

### CP3.1 · DESIGN

| | |
|---|---|
| Universe | 8 ETFs + BIL. BCOMTR index for BCI over the whole sample (BCI listed 2017). |
| Weight rule | Basket method 50/25/25, ERC inside baskets, covariance of the last 60 months. |
| Rebalancing | Monthly, to the SAA weights; rest in BIL; no transaction costs. |
| Sample | Jul 2013 – Sep 2026, 159 monthly total returns (USD, Bloomberg); T-bill: FRED. |

### CP3.2 · VALIDATION

| | |
|---|---|
| Tuning | Oct 2021 – Sep 2026 (60 months): the window that sets the weights. |
| Hold-out | Jul 2013 – Sep 2021 (99 months), never used for the weights. The split test uses only past data. |
| Variations | 37: 11 allocations, each backtested in 3 designs (full sample, hold-out, split); 23 variants compared ex ante on earlier universes; 3 changes of universe or rule. |

The 3 changes were STIP → TIP, VT → VTI + VEA + IEMG and 1/σ → ERC in the bond basket. Only the four basket budgets were candidates; equal weight and risk parity 1/σ were the Checkpoint 2 benchmarks; minimum variance, maximum Sharpe, mean–variance at 8%, ERC on 8 ETFs and maximum diversification were diagnostics. In the hold-out, risk parity 1/σ beat our allocation (0.78 against 0.66) and we did not adopt it. The hold-out is not fully clean: we saw its Sharpe Ratios when choosing the budget, though the 8% target decided (Class 4: alternating tuning and validation contaminates the validation data).

### CP3.3 · RISK

| | TUNING SAMPLE | HOLD-OUT SAMPLE | FULL SAMPLE |
|---|---|---|---|
| Mean excess return, annualised (t-stat.) | 3.15% (0.87) | 4.61% (1.91) | 4.06% (2.00) |
| Volatility, annualised | 8.13% | 6.94% | 7.39% |
| Maximum drawdown | -13.04% | -11.24% | -13.04% |
| One-month 5% VaR | 3.86% | 3.30% | 3.51% |
| One-month 5% VaR, USD | 3,860,471 | 3,295,225 | 3,508,981 |
| Sharpe Ratio | 0.39 | 0.66 | 0.55 |

*Fixed SAA weights. VaR = 1.645 × realised monthly volatility (normal). Cumulative returns: Figure 2.*

### CP3.4 · THE DISCOUNT

BACKTESTED SHARPE (HOLD-OUT)

**0.66**

REALISTIC (DISCOUNTED)

**0.33**

We start from the hold-out, the only test of our weights on months they were not estimated on, and halve it (Class 4 rule of thumb). A raw 0.66 would overstate what we expect, for five reasons. (1) With 37 variations, some look good by chance alone. (2) The hold-out was consulted when we chose the budget. (3) The ETFs were chosen in 2026, knowing that gold had rallied. (4) In 2013–2021 rates fell and stocks and bonds hedged each other, which no longer holds. (5) There are no trading costs. The t-statistic of 1.9 also means a standard error of about 0.35 on the Sharpe Ratio. We do not discount more because our weights have a single tuned parameter, the equity budget, and use no expected returns. This 0.33 is consistent with evidence that does not depend on our choices: in Class 3's 40-year horse race, risk parity earned 0.53 out of sample and a 60/40 portfolio 0.36.

# The code

All the numbers in this report come from one Python notebook, attached to the report and published on our website. Running it from top to bottom reproduces every table.

| | |
|---|---|
| `SAA_R1_our_version.ipynb` | §1–2 prices, T-bill and monthly returns; §3 the ETFs; §4 inputs on the last 60 months; §5 volatility, risk contributions and VaR; §6–7 the eleven allocations, ours included; §8 three backtests (full sample, hold-out, split); checkpoint cells for Checkpoints 1–3; §9 the shares for Bloomberg; §10 the other figures of the report. |
| `AMC_adjusted_prices_final.csv` | Bloomberg export: daily adjusted prices (dividends and splits included) of the eight ETFs, BIL and the BCOMTR index, January 2012 – 15 October 2026. |
| `AMC_closing_raw.csv` | Bloomberg export: raw closes of Friday 16 October 2026 of the eight ETFs and BIL, BCI included, used by §9. |
| `FRED DGS1MO` | Daily 1-month T-bill rate, downloaded by the notebook at every run (copy saved in DGS1MO.csv) and averaged by month. |
| `portfolio_for_PRTU.csv` | Output of §9: the 8 positions of the table below, entered in Bloomberg (PRTU). |
| `report_figures.json` | Output of §10: every number of this report, as printed here. |

# Portfolio description

Shares at $P_i$, the close of Friday 16 October 2026, in three steps. Bonds and commodities first: $N_i = \lfloor w_i \times \text{USD } 100 \text{ million} / P_i \rfloor$. Equity then takes all the capital left (the optimiser's 0.01% residual and the rounding leftovers), split among VTI, VEA and IEMG by weight, rounded down. The last USD 144.20 buys two more VEA shares, the cheapest equity ETF. No BIL: our ex-ante volatility is below the 9% cap. One purchase at that close, entered in Bloomberg PRTU on 17–18 October, unchanged until Round 2.

| SECURITY | BLOOMBERG TICKER | CCY | SHARES | PRICE | WEIGHT |
|---|---|---|---|---|---|
| Vanguard Total Stock Market ETF | `VTI US Equity` | USD | 29,149 | 374.07 | 10.90% |
| Vanguard FTSE Developed Markets ETF | `VEA US Equity` | USD | 147,369 | 69.49 | 10.24% |
| iShares Core MSCI Emerging Markets ETF | `IEMG US Equity` | USD | 133,320 | 76.79 | 10.24% |
| Vanguard Total Bond Market ETF | `BND US Equity` | USD | 188,225 | 70.82 | 13.33% |
| Vanguard Total International Bond ETF | `BNDX US Equity` | USD | 349,731 | 47.13 | 16.48% |
| iShares TIPS Bond ETF | `TIP US Equity` | USD | 144,602 | 105.54 | 15.26% |
| SPDR Gold Shares | `GLD US Equity` | USD | 29,177 | 382.01 | 11.15% |
| abrdn Bloomberg All Commodity Strategy K-1 Free ETF | `BCI US Equity` | USD | 469,258 | 26.42 | 12.40% |
| Cash, rounding leftover (USD 5.22) | | USD | | | 0.00% |
| **Total (USD 100,000,000)** | | **USD** | | | **100.00%** |

*Price date: close of Friday 16 October 2026 for every position. Actual weights after rounding: at most 0.004 percentage points from the targets of section 06 (equity slightly above). Portfolio held: ex-ante volatility 8.13%, 1-month 5% VaR 3.86% = USD 3,861,174.*

# Sources


[1] Bloomberg ECFC, consensus economic forecasts (US, developed economies, EU, emerging economies).

[2] Bloomberg WIRP, fed funds futures implied rates, priced 12 Oct 2026.

[3] U.S. Department of the Treasury, Daily Treasury Par Yield Curve and Real Yield Curve Rates, 9 Oct 2026, 9 Oct 2025 and 8 Oct 2025.
https://home.treasury.gov/resource-center/data-chart-center/interest-rates

[4] Federal Reserve Bank of St. Louis, FRED: T5YIE, T10YIE, DFII10, BAMLC0A0CM, BAMLH0A0HYM2 (ICE BofA), VIXCLS, DTWEXBGS, retrieved 12 Oct 2026.
https://fred.stlouisfed.org

[5] Vanguard, Vanguard Morningstar Total Stock Market ETF (VTI) fact sheet, 30 Sep 2026.
https://workplace.vanguard.com/iippdf/pdfs/FS970R.pdf

[6] MSCI, MSCI ACWI Index (USD) factsheet, 30 Sep 2026 (United States 64.21%).
https://www.msci.com/documents/10199/255599/msci-acwi-net.pdf

[7] FTSE Group, Country Classification Update 2008 (South Korea promoted to Developed, effective Sep 2009).
https://www.lseg.com/content/dam/ftse-russell/en_us/documents/country-classification/ftse-country-classification-update-2008.pdf

[8] The Korea Times, “Seoul to continue push for MSCI developed market status after remaining in emerging category”, 24 Jun 2026.
https://www.koreatimes.co.kr/economy/20260624/seoul-to-continue-push-for-msci-developed-market-status-bid-after-remaining-in-emerging-category

[9] World Gold Council, Gold Demand Trends Q1 2026 (central-bank purchases by quarter; Q1 2026 revised from 244t to 57t; LBMA average USD 4,873/oz).
https://www.gold.org/goldhub/research/gold-demand-trends/gold-demand-trends-q1-2026

[10] Team research, ETF_List_full.xlsx and ETF_Macro_Research_2026.xlsx (inception dates, index descriptions, DBC energy share and TER); issuer fact sheets; to be verified on Bloomberg DES.

[11] Bloomberg, Bloomberg Commodity Index Methodology, Feb 2026 (diversification limits).
https://assets.bbhub.io/professional/sites/27/BCOM-Methodology-FEB-2026.pdf

[12] Historical Option Data, market report, October 2025 (VIX close, 8 Oct 2025); to be verified on Bloomberg (VIX Index HP) or FRED VIXCLS.
https://historicaloptiondata.com/2025/10/page/22/

[13] Bloomberg, daily adjusted prices (dividends and splits included) of VTI, VEA, IEMG, BND, BNDX, TIP, GLD, BIL and the BCOMTR Index, Jan 2012 – 15 Oct 2026, and raw closes of 16 Oct 2026 of the eight ETFs and BIL, with BCI US Equity; files AMC_adjusted_prices_final.csv and AMC_closing_raw.csv, attached with the code.

[14] Federal Reserve Bank of St. Louis, FRED: DGS1MO, Market Yield on U.S. Treasury Securities at 1-Month Constant Maturity, downloaded by the notebook at each run (copy saved in DGS1MO.csv).
https://fred.stlouisfed.org/series/DGS1MO

[15] Maillard, S., Roncalli, T. and Teïletche, J. (2010), “The Properties of Equally Weighted Risk Contribution Portfolios”, Journal of Portfolio Management 36(4), 60–70.

[16] Dimson, E., Marsh, P. and Staunton, M. (2002), Triumph of the Optimists: 101 Years of Global Investment Returns, Princeton University Press.

[17] Kahn, J. (2025), “Nvidia’s $100 billion investment in OpenAI has analysts asking about ‘circular financing’ inflating an AI bubble”, Fortune, 28 Sep 2025.
https://fortune.com/2025/09/28/nvidia-openai-circular-financing-ai-bubble/

[18] Bloomberg, Bloomberg Commodity Index 2026 target weights, press release, 30 Oct 2025.
https://www.prnewswire.com/news-releases/bloomberg-commodity-index-2026-target-weights-announced-302600263.html

# The team

| | | | |
|---|---|---|---|
| Luca Di Pietro | Analyst | Tommaso Rotondò | Analyst |
| Filippo Selmi | Analyst | Giulia Musi | Analyst |
| Elena Rocco | Analyst | Aaron Arauz | Analyst |

## METHODOLOGY

Returns are monthly total returns in USD, before transaction costs. Round 1 is measured against its 8% volatility target; there is no common benchmark. The Sharpe Ratio is computed on returns in excess of the 1-month T-bill.

Matterhorn Capital Management
Università della Svizzera italiana · Via Giuseppe Buffi 13, Lugano · team@matterhorn.capital

Matterhorn Capital Management is a simulated portfolio managed by students for the USI Asset Management Challenge. All figures are based on virtual capital and are presented for educational purposes only. Past performance is not indicative of future results. Nothing in this report constitutes investment advice or an offer to buy or sell any security.